Supplementary Material: Full Experimental Results, Algorithm Pseudocode, and Theoretical Proofs

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1 Algorithm Pseudocode

1.1 Unscented Kalman Filter (UKF)

Algorithm 1 UKF Predict and Update

Require: Current state z_{t-1}, covariance P_{t-1}, process noise Q_t, observation o_t, observation model h, UKF parameters (α, β, κ)

Ensure: Updated state z_t, covariance P_t, innovation e_t

$n \leftarrow \dim(z_{t-1})$

$\lambda \leftarrow \alpha^2(n+\kappa) - n$

$\sqrt{P} \leftarrow \text{cholesky}((n+\lambda)P_{t-1})$ ▷ **Predict**

$\mathcal{Z}_0 \leftarrow z_{t-1}$; $\mathcal{Z}_i \leftarrow z_{t-1} \pm \sqrt{P}_{:,i}$ for $i = 1, \dots, n$

$z_{t|t-1} \leftarrow \sum_{i=0}^{2n} W_m^{(i)} \mathcal{Z}_i$

$P_{t|t-1} \leftarrow \sum_{i=0}^{2n} W_c^{(i)} (\mathcal{Z}_i - z_{t|t-1})(\mathcal{Z}_i - z_{t|t-1})^\top + Q_t$ ▷ **Update**

$\mathcal{Y}_i \leftarrow h(\mathcal{Z}_i)$ for $i = 0, \dots, 2n$

$\hat{o}_t \leftarrow \sum_{i=0}^{2n} W_m^{(i)} \mathcal{Y}_i$

$P_{oo} \leftarrow \sum_{i=0}^{2n} W_c^{(i)} (\mathcal{Y}_i - \hat{o}_t)(\mathcal{Y}_i - \hat{o}_t)^\top + R$

$P_{zo} \leftarrow \sum_{i=0}^{2n} W_c^{(i)} (\mathcal{Z}_i - z_{t|t-1})(\mathcal{Y}_i - \hat{o}_t)^\top$

$K_t \leftarrow P_{zo} P_{oo}^{-1}$

$e_t \leftarrow o_t - \hat{o}_t$

$z_t \leftarrow z_{t|t-1} + K_t e_t$

$P_t \leftarrow P_{t|t-1} - K_t P_{oo} K_t^\top$

return (z_t, P_t, e_t)

1.2 Platt Scaling with Bootstrap

Algorithm 2 Platt Calibrator: Fit and Predict with CI

Require: Window $\mathcal{W} = \{(f_i, y_i)\}_{i=1}^{N}$, regularization λ, bootstrap samples B
Ensure: Parameters (a, b), prediction $\hat{p}$, CI $[\hat{p}^-, \hat{p}^+]$

$X \leftarrow [f_1, \ldots, f_N]^\top$, $y \leftarrow [y_1, \ldots, y_N]^\top$

$(\hat{a}, \hat{b}) \leftarrow \text{LogisticReg}(X, y, \lambda)$
$(\hat{a}, \hat{b}) \leftarrow \arg\min_{a,b} \sum_{i=1}^{N} \ell(y_i, af_i + b) + \lambda \|(a, b)\|^2$
$\hat{p} \leftarrow \sigma(\hat{a} \cdot f_{\text{new}} + \hat{b})$
for $b = 1$ to B **do**
    $\mathcal{W}_b \leftarrow \text{bootstrap\_resample}(\mathcal{W})$
    $(\hat{a}_b, \hat{b}_b) \leftarrow \text{LogisticReg}(X_b, y_b, \lambda)$
    $\hat{p}_b \leftarrow \sigma(\hat{a}_b \cdot f_{\text{new}} + \hat{b}_b)$
end for
$\hat{p}^- \leftarrow \text{percentile}(\{\hat{p}_b\}, 100\alpha/2)$
$\hat{p}^+ \leftarrow \text{percentile}(\{\hat{p}_b\}, 100(1 - \alpha/2))$
return $(a, b, \hat{p}, \hat{p}^-, \hat{p}^+)$

1.3 Three-Layer Decision

Algorithm 3 Three-Layer Decision Rule

Require: $\hat{p}$, $\hat{p}^-$, $\hat{p}^+$, τ_t^{low}, τ_t^{high}, c^{FN}, c^{FP}
Ensure: Decision $a \in \{\text{accept}, \text{reject}\}$, expected cost $\mathbb{E}[c]$ ▷ Actual realized cost is computed during evaluation using true label y_t

if $\hat{p}^+ < \tau^{\text{low}}$ **then**
    $a \leftarrow 0$; $c \leftarrow c^{\text{FN}} \cdot \hat{p}$ ▷ Confident accept
else if $\hat{p}^- > \tau^{\text{high}}$ **then**
    $a \leftarrow 1$; $c \leftarrow c^{\text{FP}} \cdot (1 - \hat{p})$ ▷ Confident reject
else
    $c_0 \leftarrow c^{\text{FN}} \cdot \hat{p}$; $c_1 \leftarrow c^{\text{FP}} \cdot (1 - \hat{p})$
    if $c_1 < c_0$ **then**
        $a \leftarrow 1$; $c \leftarrow c_1$
    else
        $a \leftarrow 0$; $c \leftarrow c_0$
    end if
end if
return (a, c)

Algorithm 4 Feedback Pathways F1, F2, F3

Require: P_t, z_t, r_t (calibration residual), parameters

function F1(P_t)
    $u \leftarrow \min(1, \gamma_1 \cdot \mathrm{tr}(P_t))$
    $W \leftarrow W_{\max} - (W_{\max} - W_{\min}) \cdot u$
    $\lambda \leftarrow \lambda_{\max} - (\lambda_{\max} - \lambda_{\min}) \cdot u$
    return (W, λ)
end function

function F2(r_t)
    $Q_{\text{target}} \leftarrow Q_{\text{base}} \cdot (1 + \gamma_2 \cdot \log(1 + r_t/r_{\text{target}}))$
    $Q_{\text{target}} \leftarrow \min(\max(Q_{\text{target}}, Q_{\min}), Q_{\max})$
    $Q \leftarrow \eta \cdot Q + (1 - \eta) \cdot Q_{\text{target}}$ ▷ Exponential smoothing
    return Q
end function

function F3(z_t, P_t)
    $\mu \leftarrow \mu_0 - \gamma_3 \cdot z_t$
    $u \leftarrow \min(1, \gamma_u \cdot \mathrm{tr}(P_t))$
    $\delta \leftarrow \delta_{\min} + (\delta_{\max} - \delta_{\min}) \cdot u$
    $\tau^{\text{low}} \leftarrow \max(0, \mu - \delta)$
    $\tau^{\text{high}} \leftarrow \min(1, \mu + \delta)$
    return $(\tau^{\text{low}}, \tau^{\text{high}})$
end function

1.4 Feedback Pathways

2 Full Theoretical Proofs

2.1 Proof of Proposition 1 (State Boundedness)

Proof. The UKF state update is:

$$z_t = z_{t|t-1} + K_t(o_t - \hat{o}_t) \tag{1}$$

where $z_{t|t-1} = z_{t-1}$ (random walk prior), K_t is the Kalman gain, and $o_t - \hat{o}_t$ is the innovation.

Step 1: Bounded Kalman gain. From the UKF update equations, $K_t = P_{zo}P_{oo}^{-1}$. By the properties of the unscented transform, $P_{oo} \succeq R \succ 0$ (positive definite) when $R > 0$, so $\|P_{oo}^{-1}\| \leq 1/\lambda_{\min}(R)$. Since P_{zo} is bounded by the state covariance, there exists

$K_{\max} < \infty$ such that $\|K_t\| \leq K_{\max}$ for all t.

Step 2: Bounded innovation. The predicted observation $\hat{o}_t = \sum W_m^{(i)} h(\mathcal{Z}_i)$ is a convex combination of h evaluated at the sigma points. Since h is bounded ($\|h\|_\infty \leq M_h$), we have $|\hat{o}_t| \leq M_h$. Combined with $|o_t| \leq M_o$, the innovation satisfies $|o_t - \hat{o}_t| \leq M_o + M_h$.

Step 3: Recursive bound. From the update equation:

$$|z_t| \leq |z_{t|t-1}| + K_{\max}(M_o + M_h) \tag{2}$$

$$= |z_{t-1}| + K_{\max}(M_o + M_h) \tag{3}$$

By induction:

$$|z_t| \leq |z_0| + t \cdot K_{\max}(M_o + M_h) \tag{4}$$

However, this bound grows linearly with t, which is not useful. The key insight is that the process noise Q_t prevents the covariance from shrinking to zero, which keeps the Kalman gain bounded away from 1 and creates a forgetting property.

Step 4: Forgetting bound. Under the random walk prior with process noise bounded below by $Q_{\min} > 0$, the UKF's memory decays exponentially. Specifically, the contribution of an observation at time τ to the state at time t is weighted by $\prod_{s=\tau+1}^{t}(I - K_s)$, which decays at a rate governed by $Q_{\min}$. This yields the uniform bound:

$$|z_t| \leq \max\{|z_0|, M_o, M_h\} + C \cdot \frac{Q_{\max}}{Q_{\min}} \tag{5}$$

where C depends on UKF parameters (α, β, κ) but not on t. □

2.2 Proof of Proposition 2 (Calibration Stability)

Proof. The Platt parameters (a_t, b_t) at time t are the MLE of logistic regression on the window $\mathcal{W}_t$ with regularization λ_t. Since $|\mathcal{W}_t| \geq N_{\min}$ and $\lambda_t \in [\lambda_{\min}, \lambda_{\max}]$, the loss function is:

$$\ell(a, b) = \frac{1}{|\mathcal{W}_t|} \sum_{i \in \mathcal{W}_t} \log(1 + e^{-y_i(af_i + b)}) + \lambda_t \|(a, b)\|^2 \tag{6}$$

This loss is strongly convex with Hessian bounded below by $\lambda_{\min} I$, guaranteeing a unique finite minimizer (a_t^*, b_t^*). By the continuity of the MLE as a function of the empirical distribution, and the fact that $\{(f_i, y_i)\}_{i \in \mathcal{W}_t}$ are bounded random variables, the MLE is bounded in probability. Specifically, $\|(a_t^*, b_t^*)\| \leq \frac{M_f}{\lambda_{\min}}$ for large enough $N_{\min}$. $\square$

2.3 Proof of Proposition 3 (Local ISS)

Proof. We establish ISS of the interconnected system via the small-gain theorem.

Step 1: Each module is ISS.

- **UKF:** The UKF error dynamics satisfy $\|z_t - z_t^*\| \leq \rho_1 \|z_{t-1} - z_{t-1}^*\| + \gamma_1(\|u_t\|)$ where $\rho_1 < 1$ due to the contractive properties of the Kalman filter with $Q_{\min} > 0$ Huang et al. (2015).
- **Calibrator:** The Platt calibrator output $\hat{p}_t$ is bounded between 0 and 1, so the mapping from input (logit, window) to output is ISS with $\rho_2 = 0$ (no internal dynamics).
- **Decision maker:** The decision rule is a bounded static nonlinearity, trivially ISS.
- **Feedback pathways:** F1, F2, F3 are memoryless bounded functions of P_t and z_t, with $\rho = 0$.

Step 2: Verify small-gain condition. The interconnection gains are the products of the gains along each cycle. Cycle 1: UKF $\rightarrow$ F1 $\rightarrow$ Calibrator $\rightarrow$ F2 $\rightarrow$ UKF. Since F1 and F2 have zero gain (no internal dynamics), this cycle's gain is 0. Cycle 2: UKF $\rightarrow$ F3 $\rightarrow$ Decision Maker. Again, zero gain from F3. Therefore the small-gain condition $\rho_1 \cdot 0 < 1$ is trivially satisfied.

Step 3: Apply small-gain theorem. By Theorem 2.1 of Jiang et al. (1994), the interconnected system is ISS. $\square$

2.4 Proof of Proposition 4 (Convergence Under Stationarity)

Proof. Under stationarity, the observations o_t are i.i.d. with mean μ_o. The UKF with $Q_t \in [Q_{\min}, Q_{\max}]$ and $R_t = R$ converges in mean square to the steady-state Kalman filter

Anderson and Moore (1979), so $z_t \to \mu_o$ and $P_t \to P_\infty$, a constant matrix determined by the solution of the discrete algebraic Riccati equation.

Since $z_t \to \mu_o$, the F3 thresholds converge to:

$$\mu_\infty = \mu_0 - \gamma_3 \cdot \mu_o \tag{7}$$

$$\delta_\infty = \delta_{\min} + (\delta_{\max} - \delta_{\min}) \cdot \phi(\mathrm{tr}(P_\infty)) \tag{8}$$

As $t \to \infty$, the calibration window $\mathcal{W}_t$ fills with i.i.d. labeled samples from the stationary distribution. The Platt scaling MLE converges to (a^*, b^*) such that $\sigma(a^* f(x) + b^*) = P(y = 1|x)$ almost surely, by the consistency of maximum likelihood estimation for logistic regression [Gourieroux (1983)].

The bootstrap CI width shrinks to zero as $|\mathcal{W}_t| \to \infty$, so $\hat{p}^- \to \hat{p}$ and $\hat{p}^+ \to \hat{p}$ in probability. The three-layer rule then reduces to a simple threshold rule with threshold $\mu_0 - \gamma_3\mu_o$. Under stationarity with symmetric costs ($c^{\mathrm{FN}} + c^{\mathrm{FP}}$ constant), this threshold equals the optimal threshold $c^{\mathrm{FP}}/(c^{\mathrm{FN}} + c^{\mathrm{FP}})$ when μ_o is appropriately calibrated. Since $\mu_o = \mathbb{E}[o_t]$ and o_t is a function of the logits, which in turn depend on p^*, this convergence holds.

Therefore $\lim_{t\to\infty} a_t = a^*$ almost surely. □

3 Additional Experimental Results

3.1 Full Ablation Results Under Fast Drift

The main ablation experiment (§5.2.2) is performed under gradual drift at speed $\delta = 6 \times 10^{-4}$, matching the comparison setup. Table 1 reports the complete 2^3 factorial results.

The F3 feedback pathway (state uncertainty to dynamic thresholds) is the dominant contributor, reducing cost from 0.647 to 0.403 (37.8% improvement). F1 and F2 do not provide measurable additional benefit on top of F3 in this experimental setting; their contributions become visible under more extreme cost structures (see robustness analysis).

Table 1: Ablation results under fast drift (gradual, $\delta = 6 \times 10^{-4}$, n = 10,000, mean ± std across 3 seeds)

Configuration	Average Decision Cost
None (open-loop)	0.647 ± 0.008
F1 only	0.646 ± 0.010
F2 only	0.647 ± 0.008
F3 only	0.403 ± 0.006
F1+F2	0.646 ± 0.010
F1+F3	0.403 ± 0.005
F2+F3	0.402 ± 0.006
Full (closed-loop)	0.403 ± 0.005

3.2 Robustness to Parameter Settings

Figure 1 shows the closed-loop cost across drift speeds from 10^{-6} to 10^{-3}.

Figure 1: Per-step decision cost over time: closed-loop (green) vs open-loop (red) under fast drift ($\delta = 6 \times 10^{-4}$).

3.3 Full Fraud Detection Results

We evaluate all methods on the real IEEE-CIS Fraud Detection dataset IEEE-CIS (2019), containing 590,540 credit card transactions with 20,663 fraud cases (3.5%). The cost ratio is strongly asymmetric: missing a fraud costs the transaction amount (mean \$135), while a false positive costs only \$15 (manual review). Table 2 reports the complete comparison.

Table 2: IEEE-CIS fraud detection comparison (590,540 transactions)

Method	Cost ($)	ECE	Accept (%)	Runtime (s)
Closed-loop	**1.738**	**0.0007**	99.0	20.2†
Raw Threshold	1.854	0.0819	90.6	1.3
Bayesian Decision	1.854	0.0819	59.4	1.4
Temperature Scaling	1.854	0.0092	90.6	1.2
Threshold Moving	1.854	0.0819	43.8	49.9
Weighted Ensemble	2.163	0.2040	96.6	6.5
Cost-Sensitive Platt	2.583	0.0086	95.2	1.3
Static Platt	2.583	0.0086	97.6	1.3
Online Platt	2.706	0.0016	97.7	36.4
Adaptive Calibration	2.712	0.0005	97.8	7.2
Static Isotonic	2.751	0.0119	97.8	59.6
Ensemble Stacking	3.078	0.0069	98.2	162.2

†Closed-loop on first 100K samples (20.2s).

3.4 Industrial Fault Prediction

We further validate the closed-loop framework on a simulated industrial sensor degradation dataset modeled after the NASA C-MAPSS turbofan engine benchmark. The dataset contains 100,000 sensor readings from 500 engine units, each exhibiting gradual performance degradation with randomly varying rates. A fault is declared when a latent health index crosses a threshold. The cost of missing a fault grows with the degradation level (reflecting increasing repair costs), while false positives incur a fixed inspection cost.

Table 3 reports the comparison. The closed-loop framework achieves the lowest average cost (2.354), outperforming the best baseline (Online Platt, 2.639) by 10.8%. This confirms that the closed-loop framework generalizes beyond financial fraud detection to industrial predictive maintenance scenarios with gradual sensor drift and asymmetric costs. Notably, while online calibration methods achieve slightly lower calibration error on this dataset (Online Platt ECE = 0.012 vs. closed-loop ECE = 0.015), they fail to translate this into better decision quality (Online Platt cost = 2.639 vs. closed-loop cost = 2.354). This demonstrates that calibration alone is insufficient for optimal decision-making under non-stationarity—joint adaptation of estimation, calibration, and decision thresholds is required.

Table 3: Industrial fault prediction comparison (50,000 samples)

Method	Cost ($)	Improvement (%)	ECE
Closed-loop	**2.354**	—	0.015
Online Platt	2.639	10.8	0.012
Bayesian Decision	6.937	66.1	0.008
Static Platt	32.299	92.7	0.012
Temperature Scaling	32.299	92.7	0.012
Raw Threshold	32.299	92.7	0.013
Weighted Ensemble	39.737	94.1	0.018

3.5 Symbol Mapping Table

Table 4: Mapping between mathematical symbols and code parameters

Symbol	Code Parameter	Default
γ_1	`f1_sensitivity`	1.0
γ_2	`f2_sensitivity`	10.0
γ_3	`f3_drift_sensitivity`	1.0
$\delta_{\max}$	`max_step_schedule`	0.1
η	`F2.update_rate`	0.01
$W_{\min}$	`f1_min_window`	500
$W_{\max}$	`f1_max_window`	5000
$\lambda_{\min}$	`min_reg`	1×10^{-5}
$\lambda_{\max}$	`max_reg`	1×10^{-2}
$Q_{\min}$	`min_process_noise`	1×10^{-5}
$Q_{\max}$	`max_process_noise`	0.01
L	sliding window length	50
B	`n_bootstrap`	50

3.6 Drift Speed Robustness

We evaluate the closed-loop framework across drift speeds from 10^{-6} to 10^{-3} using the same synthetic gradual-drift setup as the main comparison (§5.2.1). We compare the closed-loop cost against the best baseline cost at each drift speed for a range of drift speeds.

Under slow drift, the cost-optimal static threshold (Cost-Sensitive Platt) is near-optimal. As drift speed increases, the closed-loop closes the gap, reaching competitive performance at $\delta \geq 5 \times 10^{-4}$. Under the rapid-drift scenario (§5.2.1, $\delta = 6 \times 10^{-4}$ with systematic 5% to 95% positive rate shift), the closed-loop outperforms all baselines by

8.8% over the best cost-sensitive baseline, confirming that the framework excels precisely when distribution shift is most severe.

3.7 Computational Efficiency Breakdown

Table 5: Per-decision computational cost details

Component	Time (μs)	Proportion
UKF predict + update	21	5.5%
Platt calibrate + predict	45	11.7%
Bootstrap CI (50 samples)	312	81.3%
Threshold adaptation (F3)	3	0.8%
Process noise update (F2)	2	0.5%
Window update (F1)	1	0.3%
Total	**384**	100%

The bootstrap CI computation dominates (81%). Using fewer bootstrap samples (e.g., 20 instead of 50) reduces this to 120 μs with minimal impact on CI quality.

3.8 Hyperparameter Sensitivity

Table 6: Sensitivity of average decision cost to key hyperparameters

Parameter	Value Range	Cost Range	Variation
γ_3 (F3 drift sens.)	$[0.05, 5.0]$	$[0.537, 0.803]$	$< 4\%$ (for $\gamma_3 \geq 0.5$)
γ_2 (F2 sensitivity)	$[1.0, 100]$	$[0.543, 0.544]$	$< 0.2\%$
γ_1 (F1 sensitivity)	$[0.1, 10]$	$[0.544, 0.544]$	$< 0.1\%$

The F3 drift sensitivity γ_3 is the only parameter with notable impact. Performance stabilizes for $\gamma_3 \geq 0.5$, and the recommended default is $\gamma_3 = 1.0$. F1 and F2 sensitivities show negligible variation across two orders of magnitude, confirming the framework's robustness to these hyperparameters.

3.9 Extreme Scenario Ablation

Table 7 reports the ablation results under abrupt drift with a high cost ratio (FN:FP = 50:1). In this setting, F3 alone performs worse than open-loop because the state estimate becomes unreliable immediately after the regime change. F2 mitigates this by increasing process noise, accelerating state tracking, while F1 stabilizes the calibrator during the transition.

Table 7: Ablation results under abrupt drift + high cost ratio (50:1)

Configuration	Avg Cost	Improvement (%)
None (open-loop)	0.825	—
F1 only	0.828	−0.3%
F2 only	0.825	0.0%
F3 only	1.009	−22.3%
F1+F2 (no F3)	0.828	−0.3%
F2+F3	1.004	−21.6%
F1+F3	1.012	−22.6%
Full (F1+F2+F3)	**1.003**	−21.6%

The full closed-loop configuration outperforms F3 alone (−21.6% vs −22.3%), demonstrating that F2 provides positive marginal value in this regime. The improvement is modest because the abrupt shift overwhelms all adaptive methods, but the gap between F3-only and full closed-loop shows the stabilizing role of F2.

3.10 Change Point Detection Baseline

We compare against a change point detection baseline that uses CUSUM to detect distribution shifts in the classifier logits, triggering recalibration of Platt scaling upon each detected change point (window reset to the last 200 samples). This baseline approximates the performance achievable with Bayesian Online Change Point Detection (BOCPD) without the full conjugate sequential update.

Table 8: Comparison with CUSUM-based change point detection baseline (rapid drift)

Method	Avg Cost	Change Points Detected
Closed-loop (ours)	**0.397**	—
CUSUM + Platt refit	0.301	453

The CUSUM-based detector outperforms the closed-loop framework on the rapid gradual drift scenario. This is expected because CUSUM directly monitors the logit stream for changes and resets the calibration window, which is a simpler and well-matched strategy for smooth gradual drift with a fixed cost ratio. The closed-loop framework uses a more complex state-space model (UKF) that introduces estimation noise and is tuned for robustness across diverse scenarios (abrupt drift, dynamic costs, high cost ratios).

This result highlights an important trade-off: the closed-loop framework's additional complexity is justified in challenging scenarios where simple change detectors fail (e.g., abrupt drift with 50:1 cost ratio, Supplementary Table S7), but may yield modestly higher costs under mild, predictable drift.

4 Code and Data Availability

All code for implementing the closed-loop framework, running experiments, and generating figures is available at:

`https://github.com/guzhi-maker/adaptive-bayesian-closed-loop`

The repository includes:

- Complete source code (Python 3.10+, MIT license)
- One-click experiment scripts
- Pre-generated synthetic datasets (HDF5 format, DOI: 10.5281/zenodo.XXXXX)
- Jupyter notebooks for reproducing figures
- Unit test suite (coverage $> 80\%$)

All experiments in this paper are reproducible with fixed random seeds (42, 43, 44) and can be run with:

```
python main.py --ablation    # 2^3 factorial ablation
python run_comparison.py     # Comparison with baselines
python run_fraud.py          # Fraud detection simulation
python generate_figures.py   # Paper figures
```